Nihal Mohan Shettigar

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Software engineer building full-stack AI applications and enterprise automation, with hands-on experience in Python, TypeScript, backend APIs, and data pipelines.

Experience

- **Cognizant Technology Solutions** — Chennai, India
  Programmer Analyst — *June 2025 – Present*
  - Validated enterprise data during SAP ECC to S/4HANA conversion, checking data integrity across migration workflows and large-scale datasets.
  - Developed VBS automation and Excel reporting pipelines to reduce repetitive manual work in enterprise data processing and operational reporting.

Internships

- **National Institute of Technology Karnataka** — Surathkal, Mangalore
  Research Intern — *Jan 2025 – May 2025*
  - Developed a video captioning model combining computer vision and sequence-to-sequence methods to generate captions from video input.
  - Implemented an end-to-end training pipeline for frame extraction, feature encoding, and attention-based decoding for multimodal AI research.

Projects

- **CallPilot – AI Voice Assistant Simulator** — GitHub
  TypeScript, Next.js, Fastify, Supabase, Zod — *Sep 2026*
  - Built a full stack simulator for small-business missed-call follow-up, with configurable cake-order and clinic-booking workflows, conversation history, and follow-up dashboards.
  - Integrated pluggable LLM providers and schema-validated outputs with external speech-to-text and text-to-speech services, supporting English, Hindi, and mixed-language conversations.
  - Implemented Google Calendar booking, rescheduling, and cancellation with explicit confirmation; secured the integration with OAuth, encrypted credential storage, and access controls for each business.
- **CA Portal – Regulatory Intelligence Platform** — GitHub
  Next.js, React, TypeScript, Prisma, PostgreSQL, Gemini API — *Oct 2026*
  - Built automated CBIC document and RSS ingestion with PDF storage, retries, and safe processing across concurrent workers.
  - Integrated Gemini for structured extraction of regulatory summaries, key points, and statutes; built a searchable knowledge base and an editorial approval gate for AI-generated news.
  - Developed a statutory deadline engine with government extension overrides and calendar exports, alongside income-tax, TDS, HRA, and EMI calculators with Vitest unit tests.
- **Sketchify – Sketch to Realistic Face Generator**
  Python, GANs, Deep Learning — *Final Year Project | Oct 2024*
  - Trained a GAN on CelebAMask-HQ to generate realistic faces from hand-drawn sketches; built a web interface for sketch input and model inference.

Technical Skills

- **Languages**: Python, TypeScript, JavaScript, SQL
- **Backend**: FastAPI, Async Python, Fastify, Pydantic v2, REST APIs, JWT Authentication, OAuth 2.0
- **Frontend**: Next.js, React, TypeScript, Tailwind CSS, TanStack Query, Zustand
- **AI & Agents**: LangGraph, OpenAI API, Anthropic API, Gemini API, AWS Bedrock, Structured LLM Pipelines
- **Databases**: PostgreSQL, Supabase, Prisma, MongoDB, Beanie ODM, Motor (async driver)
- **Infrastructure**: Docker, Docker Compose, AWS (EC2, S3, IAM), CI/CD, GitHub Actions, Multi-stage Builds
- **Dev Tools**: Git, Claude Code, Codex, pytest, Vitest, Ruff, Mypy, Pre-commit Hooks
- **Enterprise**: SAP ECC, SAP S/4HANA, VBS Automation, Excel Automation

Education

- **NMAM Institute of Technology** — Nitte, Karnataka
  Bachelor of Engineering in Artificial Intelligence and Machine Learning; CGPA: 7.98 — *2021 – 2025*